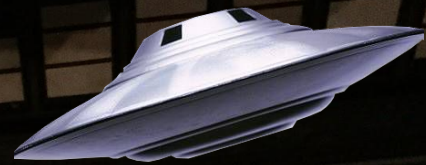


# Agile Drupal 8 Builds: Doing the Most Without PHP

Matt Cheney & Molly Byrnes



# DrupalGovcon July 20th, 2016



# Hello to Drupalcon Govcon



PANTHEON®



My name is Matthew Cheney.

I work on the magical platform that is Pantheon.

I believe in the power of site building.

I worship at the altar of Earl Miles.

I helped make Panopoly for Drupal 7.

I want Drupal to be awesome for non-coders.

I know that will happen with Drupal 8.

# Hello to DrupalGovCon



PANTHEON®



My name is Molly Byrnes.

I work at Phase2 making digital visions reality.

I believe in the promise of open technology.

I managed content on the first major corporate international Drupal platform.

I helped build one of the first large Drupal 8 sites in 2015 and a Drupal 8 media platform in 2016.



# This is a Talk is About Making Websites



- We have content.
- We want to share the content.
- We have some time to spend.
- We need some tools to do this.
- We are going to do this right.



# And the Choices We Make Along the Way



- But we have Drupal.
- Drupal let's us do a lot of things.
- Dream it? You can Drupal it.
- My dreams are pretty crazy.
- I just want to build a website.



# How about a Radical Idea for Development?

Download & Extend

[Download & Extend Home](#) [Drupal Core](#) [Distributions](#) [Modules](#) [Themes](#)

Releases

**drupal 8.1.1**

[View](#) [Revisions](#)

[Download tar.gz](#) [Download zip](#)

11.95 MB 19.5 MB





# The Value of Core

Aren't 237,000+  
lines of PHP enough?

- Complete CMS Solution
- High Quality Codebase
- Easy to Update

# This Was Not True with Drupal 7



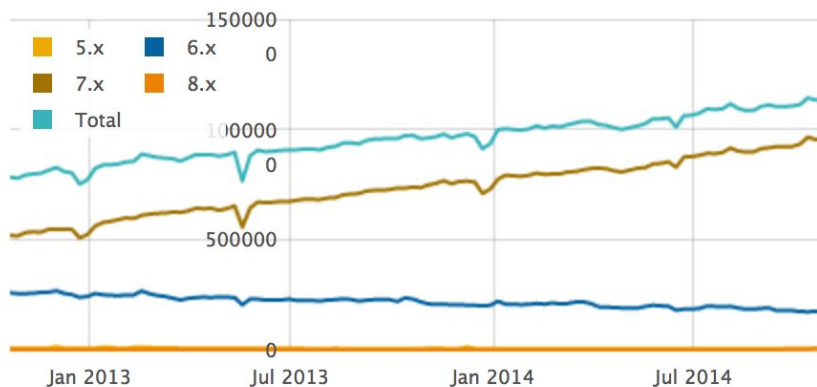
PANTHEON®



## Usage statistics for *Drupal core*

[Drupal core project page](#)  
Usage statistics for all projects

### Weekly project usage

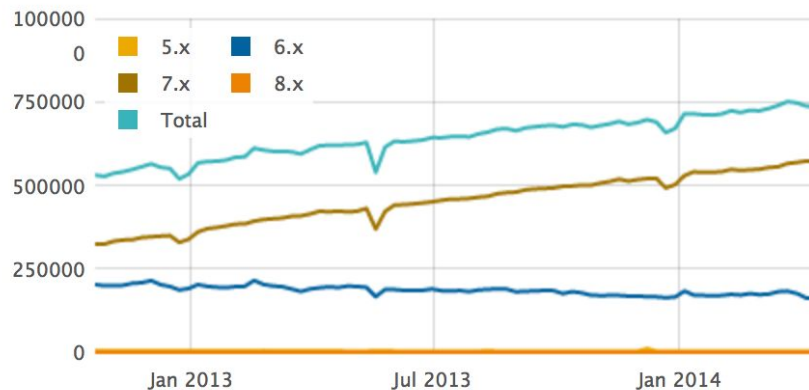


| Week           | 5.x   | 6.x     | 7.x       | 8.x    | Total     |
|----------------|-------|---------|-----------|--------|-----------|
| April 24, 2016 | 2,128 | 103,997 | 1,016,267 | 74,866 | 1,197,258 |

## Usage statistics for *Views*

[Views project page](#)  
Usage statistics for all projects

### Weekly project usage



| Week           | 5.x   | 6.x    | 7.x     | 8.x | Total   |
|----------------|-------|--------|---------|-----|---------|
| April 24, 2016 | 1,877 | 85,755 | 831,202 | 0   | 918,834 |

# And We Want Awesome Out of the Box



PANTHEON®



# I Tried to Make this Happen in Drupal



## Panopoly

[View](#) [Version control](#)

Posted by [populist](#) on March 16, 2012 at 2:16am

Panopoly is powerful base distribution of Drupal powered by lots of Chaos Tools and Panels magic. The distribution is designed to be both a general foundation for site building and a base framework upon which to build other Drupal distributions (see [Drupalcon Portland video](#)) or check out some of its key features (see full list):



- **Panels Powered** – Woven together functionality from [Panels](#), the Panels In-Place-Editor, [Fieldable Panel Panes](#), [PM Existing Pages](#), [Panelizer](#) and a lot of extra special sauce.
- **Responsive** – Over two dozen cross-browser, responsive layouts and several responsive image styles are included to aid in site building and allow your site to work seamlessly on mobile devices.
- **In-Place Page Building** – Drag and drop customization of pages and nodes. Easily switch layouts, modify Views and Panels settings, change styles, and add generic widgets to display text, images, links, maps, submenus, video and spotlights across your site. No backend administration required!
- **Admin Experience** – Vastly improved site and content management experience, featuring a redesigned node add/edit page and a fully loaded, media enabled TinyMCE WYSIWYG.
- **Apps Powered** – Panopoly can be extended through various Panopoly Apps delivered by the Panopoly App server.

To make everything happen, Panopoly bundles a [currated set of contributed modules and third party libraries](#) with its own custom modules including [Panopoly Admin](#), [Panopoly Core](#), [Panopoly Images](#), [Panopoly Magic](#), [Panopoly Pages](#), [Panopoly Search](#), [Panopoly Theme](#), [Panopoly Users](#), [Panopoly Widgets](#), and [Panopoly WYSIWYG](#).

### Maintainers for Panopoly

[dsnopek](#) – 628 commits  
last: 4 days ago, first: 2 years ago

[populist](#) – 522 commits  
last: 1 year ago, first: 4 years ago

[mrfelton](#) – 61 commits  
last: 1 year ago, first: 2 years ago

[View all committers](#)  
[View commits](#)

### Issues for Panopoly

To avoid duplicates, please search before submitting a new issue.

[Advanced search](#)

All issues

[335 open](#), [1803 total](#)

Bug report

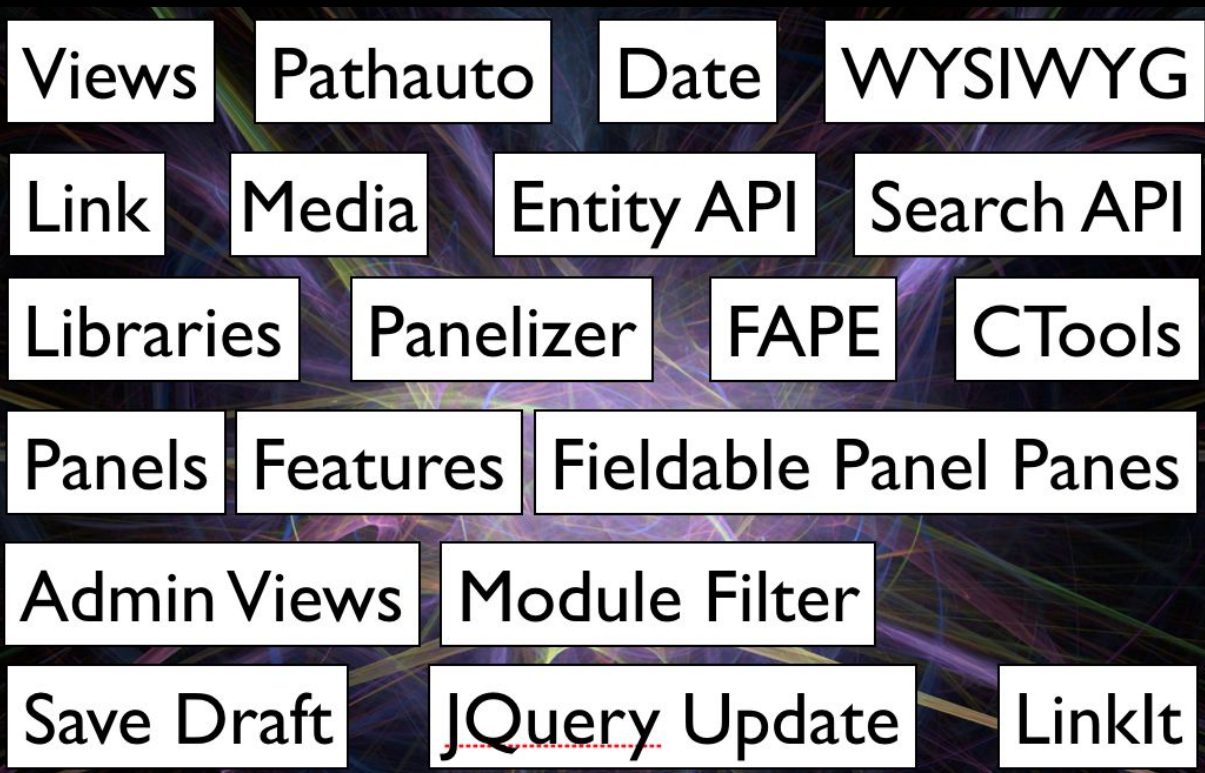
[110 open](#), [743 total](#)

[Subscribe via e-mail](#)

# Making Drupal 7 Out of the Box Awesome



# Panopoly Bundled Top Modules



# Which Gave Lots of Power to Site Builders



PANTHEON®



**My Great Panopoly Site**My accountLog out

Homepage DemoContent DemoGreat VegetablesLovely Vegetables

Home » Homepage Demo

Homepage Demo

Content

Content Demo

Garlic radicchio chaya. Mizuna greens moth bean horseradish urad bean? Gobo new zealand spinach; aubergine manioc mustard, celtuce winter purslane corn salad celeriac garbanzo.

Sidebar

Enter your keywords

Demo Items

Content Demo  
March 20, 2012

SaveCancel

14

# And Provided a WYSIWYG



PANTHEON®



**B I S** [List Icons] [Link Icon] [Image Icon] [Table Icon] [Checkmark Icon] [Print Icon] [Format] [Underline Icon] [Text Color Icon] [Background Color Icon] [Link Icon] [Unlink Icon] [Undo Icon] [Redo Icon]

Veggies sunt bona vobis, proinde vos postulo esse magis parsnip water chestnut chickweed celery earthnut pea kohlrabi courgette watercress lettuce winter purslane komatsuna.

Asparagus garbanzo arugula corn chicory collard greens salad gourd desert raisin rutabaga prairie turnip bell pepper tigernut beetroot azuki bean pea sprouts. Bamboo shoot bell pepper coriander broccoli wakame tigernut dandelion gram celery sea lettuce fennel daikon garbanzo tomato azuki bean cabbage. Celtuce bush tomato coriander eggplant peanut amaranth water spinach bunya nuts lotus root earthnut pea kombu horseradish ricebean leek squash soko.



This is a bunch of carrots. They look great!

Tatsoi cress cucumber gourd kakadu plum salsify horseradish watercress fava bean prairie turnip

Editor: WYSIWYG

# And a Ton of Other Stuff



PANTHEON®



## General Features

- **Panels Powered** – Woven together functionality from [Panels](#), the Panels In-Place-Editor, [Fieldable Panel Panes](#), [PM Existing Pages](#), [Panelizer](#) and a lot of extra special sauce.
- **Responsive** – Over two dozen cross-browser, responsive layouts and several responsive image styles are included to aid in site building and allow your site to work seamlessly on mobile devices.
- **In-Place Page Building** – Drag and drop customization of pages and nodes. Easily switch layouts, modify Views and Panels settings, change styles, and add generic widgets to display text, images, links, maps, submenus, video and spotlights across your site. No backend administration required!
- **Admin Experience** – Vastly improved site and content management experience, featuring a redesigned node add/edit page and a fully loaded, media enabled TinyMCE WYSIWYG.
- **Apps Powered** – Panopoly can be extended through various Panopoly Apps delivered by the Panopoly App server.

## Panopoly Core

- Packages a customized design for the Panels IPE experience
- Includes @font-face support for Open Sans font
- Defines a custom day and time format
- Defines "Administrative" and "Editor" user roles
- Provides a global "Categories" taxonomy with Panelizer support
- Custom styling for the CTools dropdown and jQuery UI modals
- Provides defaults about the Panopoly App Server for use by Apps.module
- Special handling of site breadcrumb to append current page title
- Installation profile specific code for cleaner profiles based on Panopoly

## Panopoly Admin

- Customized Content Editing Experience
- Custom Style for the Media Browser
- Creates Panopoly Administrative Section (admin/panopoly) for Panopoly Configuration
- Provides Pane Library, Layout Library, and Page Library Administrative Pages
- Packages Several Different Administrative Toolbar Options (Admin Menu, Admin, Navbar, Toolbar)

## Panopoly Images

- Several image style presets of various sizes
- Integration with Media.module for style presets
- Responsive CSS logic for image styles

## Panopoly Magic

- Magical customization of the Panels IPE experience
- Enables editing of specific field content in the Panels IPE
- Provides live previews of content administered using the Panels IPE
- Allows extensive customization of Views settings when added to Panels

# And a Ton of Other Stuff



**PANTHEON®**



## Panopoly Pages

- Creation of a "Content Page" content type for basic page management
- Creation of a "Landing Page" workflow for landing page creation
- Panelizer integration for all view modes of the "Content Page"

## Panopoly Search

- Integration with Search API Solr for Solr searching
- Integration with Search API for database searching
- Integration with and Panelization of Facet API module
- Panelizer search pages for DB and Solr results

## Panopoly Theme

- Includes 31 responsive and cross browser Panel layouts
- Implements "Featured" view mode available for all content types
- Accordion Panels style plugin
- Installation profile specific code to allow install profiles to select a theme

## Panopoly Users

- Panelization of the user/login, user/password and user/register pages
- Panelized user profile pages by default
- Addition of fields to support a user picture and user biography
- UI/UX Improvements to the user login and registration process

## Panopoly Widgets

- Integration with Link module to provide a list of links
- Provides the ability to provide an uploaded file
- Provides the ability to provide an uploaded image
- Provides the ability to provide a WYSIWYG authored block of text
- Integration with Simple GMap module to provide a map
- Integration with Tablefield module to provide a table
- Integration with Media module to provide an embedded video
- Provides a custom entity to create a rotating spotlight
- Integration with Menu Block to provide a submenu
- Provides a customizable view to add a generic list of content
- Provides a customizable view to add a generic piece of content
- Inclusion of default content for Panopoly Widgets
- UX/UI Improvements for naming and positioning of Chaos Tools plugins

## Panopoly WYSIWYG

- The best TinyMCE WYSIWYG in Drupal
- Addition of Kitchen Sink and Spellcheck Plugins
- A good MarkitUp Editor for Drupal
- Switching WYSIWYG + HTML text formats
- Integration of Linkit module for adding WYSIWYG Links
- Support for Caption Filter's WYSIWYG Captions

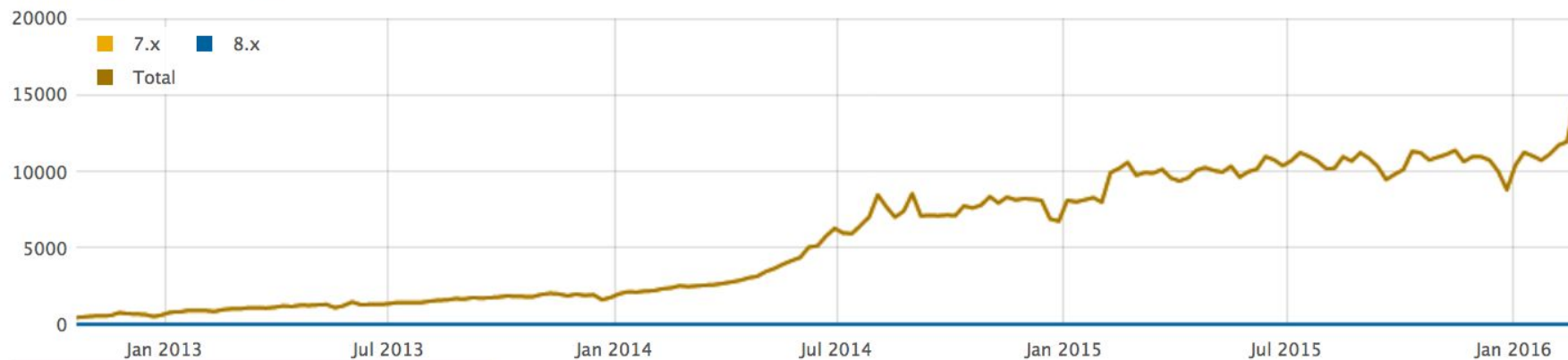
# And Panopoly Got a Lot of Love



PANTHEON®



## Weekly project usage



# But There Was a Lot of Complexity



PANTHEON®



```
panopoly_wysiwyg.info ✕  
1 name = Panopoly WYSIWYG  
2 description = Integrates a WYSIWYG into Drupal.  
3 core = 7.x  
4 package = Panopoly  
5 project = panopoly_wysiwyg  
6 dependencies[] = caption_filter  
7 dependencies[] = ctools  
8 dependencies[] = defaultconfig  
9 dependencies[] = features  
10 dependencies[] = filter  
11 dependencies[] = image_resize_filter  
12 dependencies[] = linkit  
13 dependencies[] = media  
14 dependencies[] = media_internet  
15 dependencies[] = media_vimeo  
16 dependencies[] = media_wysiwyg  
17 dependencies[] = media_youtube  
18 dependencies[] = strongarm  
19 dependencies[] = wysiwyg  
20 dependencies[] = wysiwyg_filter
```



- The more members of your core development team the better.
  - ◎ Panopoly had ~6. This limited the velocity of features/integrations.
- Defining configuration in code is important.
  - ◎ Lots of time was spent simply getting the features to work (and be extended) correctly.
- Code quality and structure vary wildly across contrib projects.
  - ◎ And can change from release to release! This makes coordinating new releases really tricky.
- Difficult to centralize contribution to the Panopoly ecosystem
  - ◎ So many different issue queues! Hard to surface issues that needed attention.

# Common Scenarios in Drupal 7 and before



- “We have no idea where this bug comes from or is inherited from in some inconsistent way”
- “Module X has a critical bug, can’t find maintainer and now I’m cowboy coding some weird patch fix”
- “There is a security hole in the theme layer because someone put an entire PHP function in it and didn’t sanitize it”
- “Deployed to production and things are weird because some feature revert override”

# Drupal 8 Gives Me Hope



PANTHEON®



# Drupal 8 Makes Much of This Better



- The more members of your core development team the better.
  - ◉ Drupal core has the largest number of core developers in the ecosystem
- Defining configuration in code is important.
  - ◉ Contribution management makes this much easier.
- Code quality and structure varies wildly across contrib projects
  - ◉ Drupal core provides clear patterns and structures for developers to leverage.
- Difficult to centralize contribution to the Panopoly ecosystem
  - ◉ <https://www.drupal.org/project/issues/drupal>



- + Major UX initiative for the administration experience
- + Brings in a lot of learnings & long term feedback from Drupal 'end users'
- + Views in core
- + Multi-lingual complete rethinking
- + Configuration management
- + Front end templating

- Templating
- More accessible to common practices
- Drupal themeing isn't so specialized
- No PHP required





## THEN..

I know that style change **seems** super small but the level of PHP I'll need to overwrite, strip and deal with because of how deep that markup is in a module file will take me an entire day to wrangle.

## NOW..

I had so much extra time on this integration that I added this sweet CSS animation on the transition between photos in this slider that I knew you would like it!

# So much MAGIC...



PANTHEON®



Integrating Pattern Library and doing exciting things are now possible

<https://www.phase2technology.com/blog/introducing-pattern-lab-starter-8/>

We have the technology:

<https://github.com/pattern-lab/edition-php-drupal-standard>

<https://github.com/phase2/pattern-lab-starter>

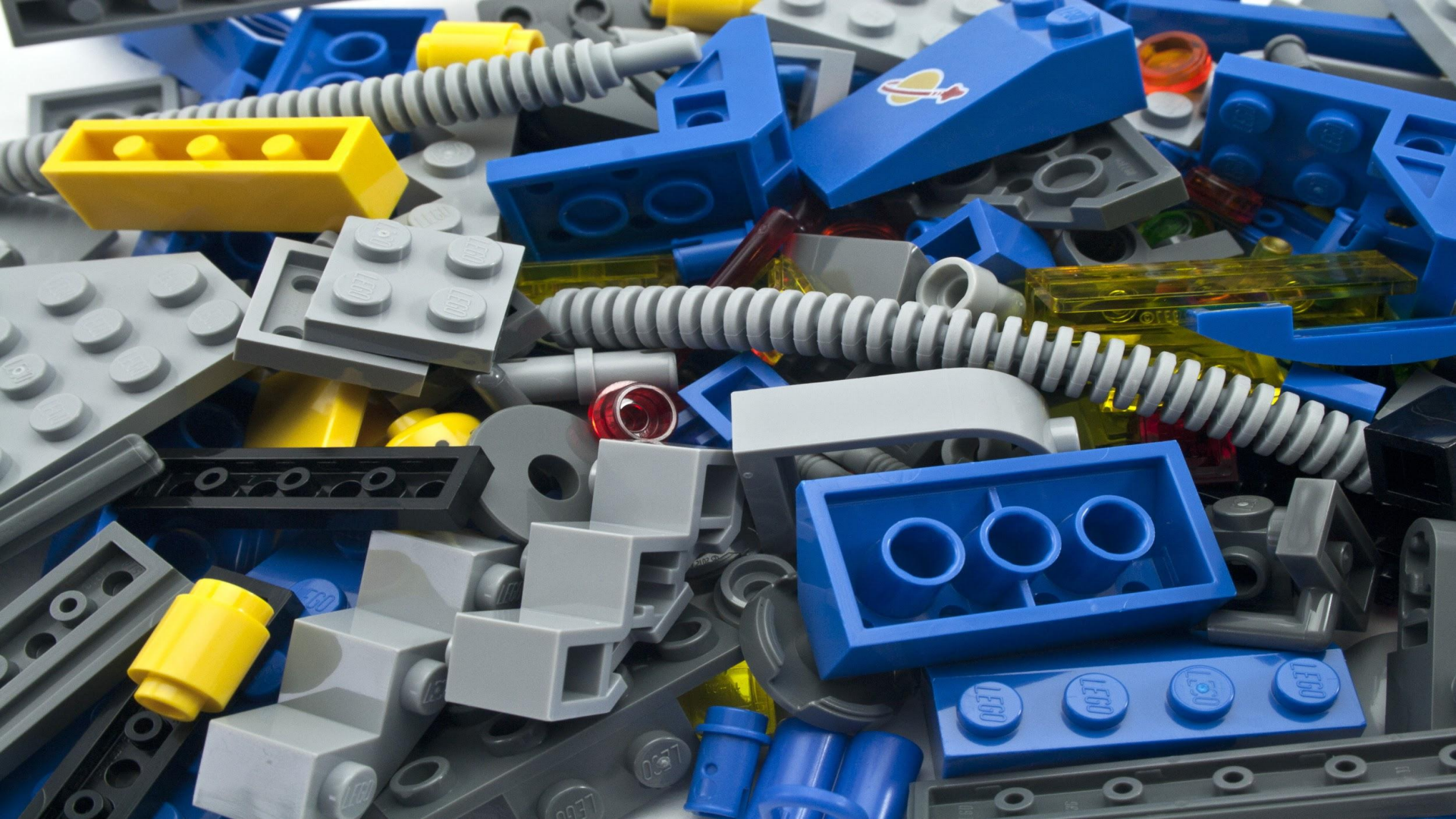
<https://github.com/phase2/drupal-lab>

# Drupal 8 Out of the Box Can Be Really Good









# Understand the contents of the box



PANTHEON®



# Help others to understand the content box



PANTHEON®



we log in to our current CMS with 3 screens, we can mirror this with the new system because it's so FLEXIBLE, right?

Let me walk you through how the default user login process works in Drupal!



Here is a list of all 120 modules on our D6 site... what is the upgrade plan for DRUPAL 8?

We won't need nearly that many and we likely accomplish a lot of that functionality with CORE. Let's talk about your content management goals first.



## Drupal 7 and before

- Content lists
- Vanity URLs
- WYSIWYG
- Configuration Management (features)

This is actually a pretty critical feature especially for folks transitioning from systems without this autonomy

## Drupal 8

- Layout options
- Social Media Logins
- Special Caching
- Workflow models
- Dynamic content tricks

## Download & Extend

[Download & Extend Home](#)[Drupal Core](#)[Distributions](#)[Modules](#)[Themes](#)[Views](#) » [Issues](#)

whhaaaHAT IS A DRUPAL?!

### Option to start Pager count at 1 instead of 0

It would be nice if it was possible to start counting pager pages from 1 instead of 0. That way the pager number will be in sync with the ?page=# number. Now it is +1 ahead.

### Comments



**dawehner** Credit commented 5 years ago

#1

Status: Active » Closed (works as designed)

This is how drupal works. See drupal.org as example.

[Log in](#) or [register](#) to post comments



**tsvenson** Credit commented 5 years ago

#2

I'm aware of that this is how Drupal works. Just hoped it was something that wasn't too complicated to add as for humans it would make more sense when looking at the URL's.

[Log in](#) or [register](#) to post comments

#### Closed (works as designed)

|             |                          |
|-------------|--------------------------|
| Project:    | Views                    |
| Version:    | 8.x-3.x-dev              |
| Component:  | Miscellaneous            |
| Priority:   | Normal                   |
| Category:   | Feature request          |
| Assigned:   | Unassigned               |
| Issue tags: |                          |
|             | VDC                      |
| Reporter:   | tsvenson                 |
| Created:    | January 12, 2011 - 13:07 |

# Drupal 8 Core won't solve everything for you



PANTHEON®



- Goals
- Requirements
- Metrics for Success
- Organizational Vocabularies
- Workflows
- Content Model
- Plan and execution strategy
- .... your organizational politics around web and digital

**GOOOOAAAALSSSSS**



**PANTHEON®**



Everything we do should be directly traceable to a goal that creates value.

**This is not as easy at it sounds...**



**PANTHEON®**



Every project or initiative starts  
someone trying to explain  
something that doesn't yet  
exist.



For any one ticket, there was an 8 step process followed, only 2 of which are development:

1. Review requirements across multiple documents
2. Study legacy system editorial experience
3. Develop initial assumptions and questions about requirements
4. Hold multiple work sessions with client to refine requirements
5. Rewrite user stories, technical approach, and acceptance criteria
6. Develop functionality
7. Export functionality into configuration
8. Test functionality



Define the acceptance criteria for a feature and then try it out on a Drupal 8 install. There might be one part of the initial ask that needs to be removed.

*System should provide user registration form with fields for username, password and email.*

*System should provide option for direct registration **or RaaS (Registration as a Service)** to manage user profile.*

*System should send email with a link to user to activate the account.*

*User should open the link in the email and activate the account for usage.*

# Content Modeling



PANTHEON®



Drupal has lots of places to put things and defining what, where and how content gets stored up front and in a systematic way leads to a smoother implementation.

|    |                 |                |                    |          |             |           |                             |
|----|-----------------|----------------|--------------------|----------|-------------|-----------|-----------------------------|
| fx | Profile         |                |                    |          |             |           |                             |
|    | A               | B              | C                  | D        | E           | F         | G                           |
| 1  | Profile         |                |                    |          |             |           |                             |
| 2  | Label           | Machine name   | Widget             | Required | # of values | Help Text | Notes                       |
| 3  | Name            | title          | Title field        | Yes      | 1           |           |                             |
| 4  | Image           | system default | Image field        | No       | 1           |           |                             |
| 5  | Bio             | body           | Body               | Yes      | 1           |           | Plain text                  |
| 6  | Department      |                | Text field         | Yes      | 1           |           |                             |
| 7  | Network         |                | Text field         | Yes      | 1           |           |                             |
| 8  | Region          |                | Taxonomy reference | Yes      | 1           |           | Reference Region vocabulary |
| 9  | Email Address   |                | Email address      | Yes      | 1           |           |                             |
| 10 | Phone Number    |                | Phone              | No       | 1           |           |                             |
| 11 | Twitter Profile |                | Link               | No       | 1           |           | Disable title               |
| 12 | Location        |                | Address field      | Yes      | 1           |           |                             |
| 13 | Interests       |                | Term reference     | Yes      | 1           |           | Reference Topics vocabulary |

# Content Modeling Benefits



PANTHEON®



- Bring together key elements of the system in one place
- Flesh out the ideal details within the structure available
- Share and distribute ownership of data structures and naming earlier in project

## Resources:

- [https://docs.google.com/spreadsheets/d/15htLLWLguhwiuTLg\\_nndQNpgWVdUMy6UaR\\_d1q-v6iw/edit#gid=0](https://docs.google.com/spreadsheets/d/15htLLWLguhwiuTLg_nndQNpgWVdUMy6UaR_d1q-v6iw/edit#gid=0)
- <https://www.palantir.net/blog/developing-drupal-sites-plan-or-perish>



Choice to using a mostly core system will require a strong commitment to this approach from all parties involved.

Feature match to the exact idea of how to achieve the end goal for user in optimal way.

Healthy Tension

Approximate match to original idea yet aligned with goal AND achievable using the flexible tools in Drupal core



agile means being able to  
move quickly and easily.



**MEASURE TWIC  
CUT ONCE**

A rectangular wooden sign with a blue border. The text "MEASURE TWIC" is on the top line and "CUT ONCE" is on the bottom line, both in a bold, serif font. The sign is placed on a dark, textured surface. In the background, there are some white and blue objects, possibly containers or tools.

BUT CUTTING IS  
MORE FUN  
THAN MEASURING

MEASURE TWICE, CUT ONCE

PEACHEY 2010

Image by Jeffery Peachy [Creative Commons Attribution-NonCommercial-NoDerivs 3.0 Unported License](https://creativecommons.org/licenses/by-nc-nd/3.0/).

<https://jeffpeachey.com/2010/12/02/measure-twice-cut-once-woodcut-for-sale/>



^ o. Define goals, figure out the requirements of what you actually need to build and align stakeholders with the approach

# The Agile D8 Build Methodology

A Better Development Process

1. Begin development with vanilla Drupal 8 core.
2. Build as much of your site as possible using Drupal 8 core.
3. Add contrib projects (or custom code) when needed.
4. Leverage power of TWIG for customizing user experience.
5. Make an awesome website

# So Let's Build Something Cool with D8



PANTHEON®



KICKSTARTER

Discover

Start a project

About us



Search Projects

Sign up

Log in

Explore the possibilities of a creative universe

15 diverse categories. Thousands of amazing projects.

Art

Comics

Crafts

Dance

Design

Fashion

Film & Video

Food

Games

Journalism

Music

Photography

Publishing

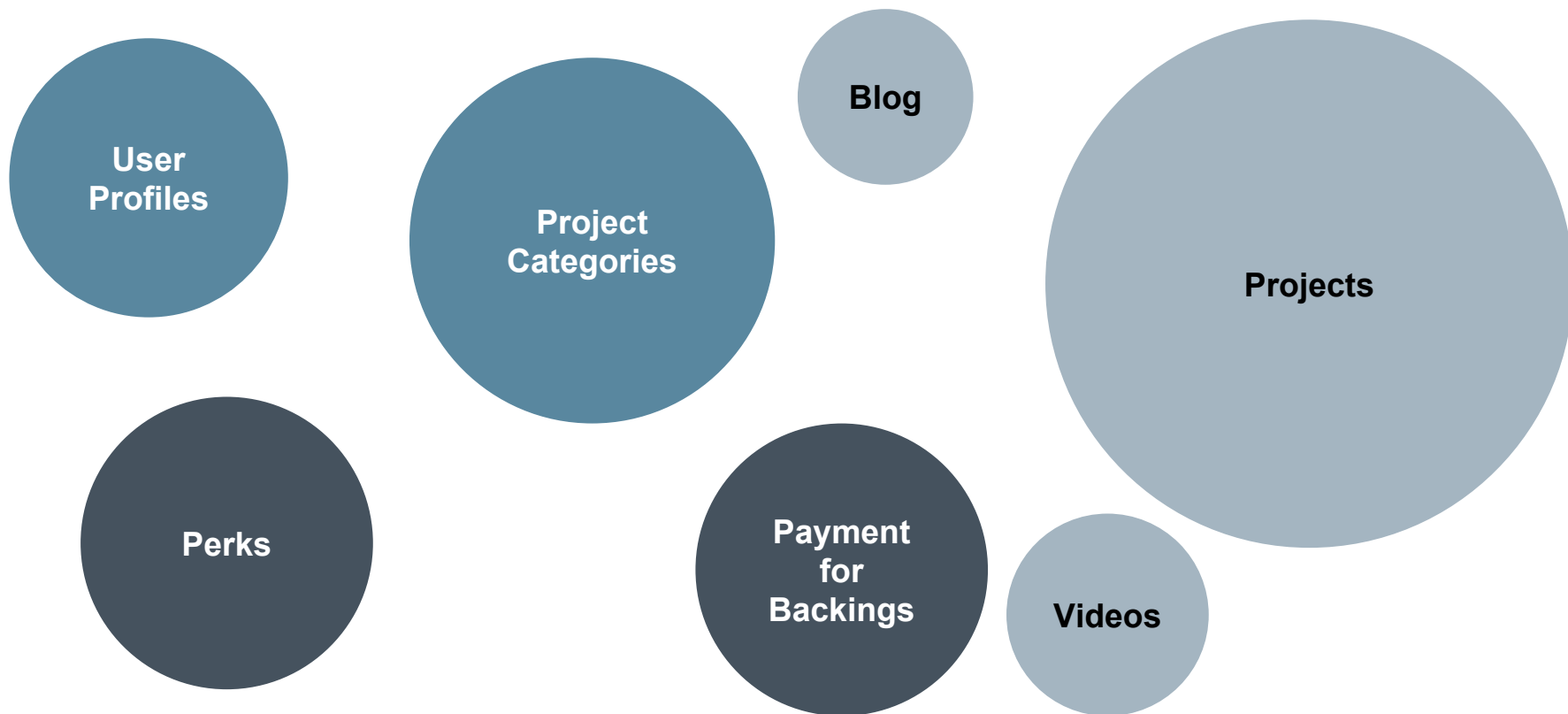
Technology

Theater

# We Had a Lot of Content to Manage



**PANTHEON®**



# And Some Complex Functionality



PANTHEON®



**User  
Signups**

**Defining and  
Fulfilling  
Perks**

**Post a  
Blog**

**Submitting a Project  
for Review**

**Project  
Search**

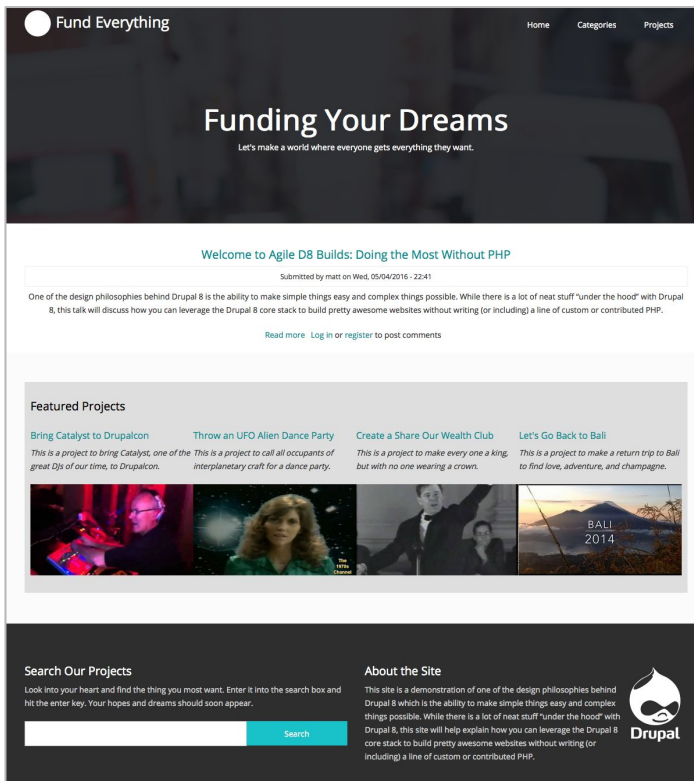
**Backing a  
Project**

**Post a  
Video**

# But We Were Able to Do Most of It With D8



PANTHEON®



- Article / Blog System
- Project Submission Workflow
- Project Categorization
- Project Perks and Fulfillment
- Project Backing & Payment
- User Accounts & Profiles
- TWIG based Drupal theme

# **Our Drupal 8 Agile Site Build**

# Start with the Basics



The screenshot shows a web browser window with the URL <https://dashboard.pantheon.io/products/drupal8/spinup>. The page has a dark header with the Pantheon logo and a user profile for "David". The main content area is split into two columns. The left column features the "Drupal 8" title, the "By: Drupal Community" text, and a blue Drupal logo. Below this, it states "Drupal 8 is the biggest update in Drupal history, and is now **officially supported on Pantheon**! Get started building with the latest Drupal on the latest stack." Further down, it says "Drupal 8 is now **officially supported on Pantheon**. Start learning about using D8 with Pantheon's workflow today." The right column is titled "Name your Drupal 8 site" and contains a form. The form has three sections: "Name Your Site" with a text input field containing "My Awesome Site" and a list of rules (less than 52 characters, only a-z, A-Z, 0-9, and dashes, cannot begin or end with a dash, cannot change); "Organization Affiliation" with a dropdown menu set to "Pantheon Support"; and "Site Development URL" with a text input field containing "dev- my-awesome-site" and a ".pantheonsite.io" suffix. A yellow "Create Site" button is at the bottom of the form. A "Help" button is in the bottom right corner of the page.

Drupal 8  
By: **Drupal Community**

Drupal 8 is the biggest update in Drupal history, and is now **officially supported on Pantheon**! Get started building with the latest Drupal on the latest stack.

Drupal 8 is now **officially supported on Pantheon**. Start learning about using D8 with Pantheon's workflow today.

### Name your Drupal 8 site

**Name Your Site**

- < 52 characters
- Can only contain a-z, A-Z, 0-9, and dashes ("-")
- Cannot begin or end with a dash
- Cannot change

**Organization Affiliation**

Pantheon Support

**Site Development URL**

dev-  .pantheonsite.io

Create Site >

? Help

# Start with Basic Drupal 8.1

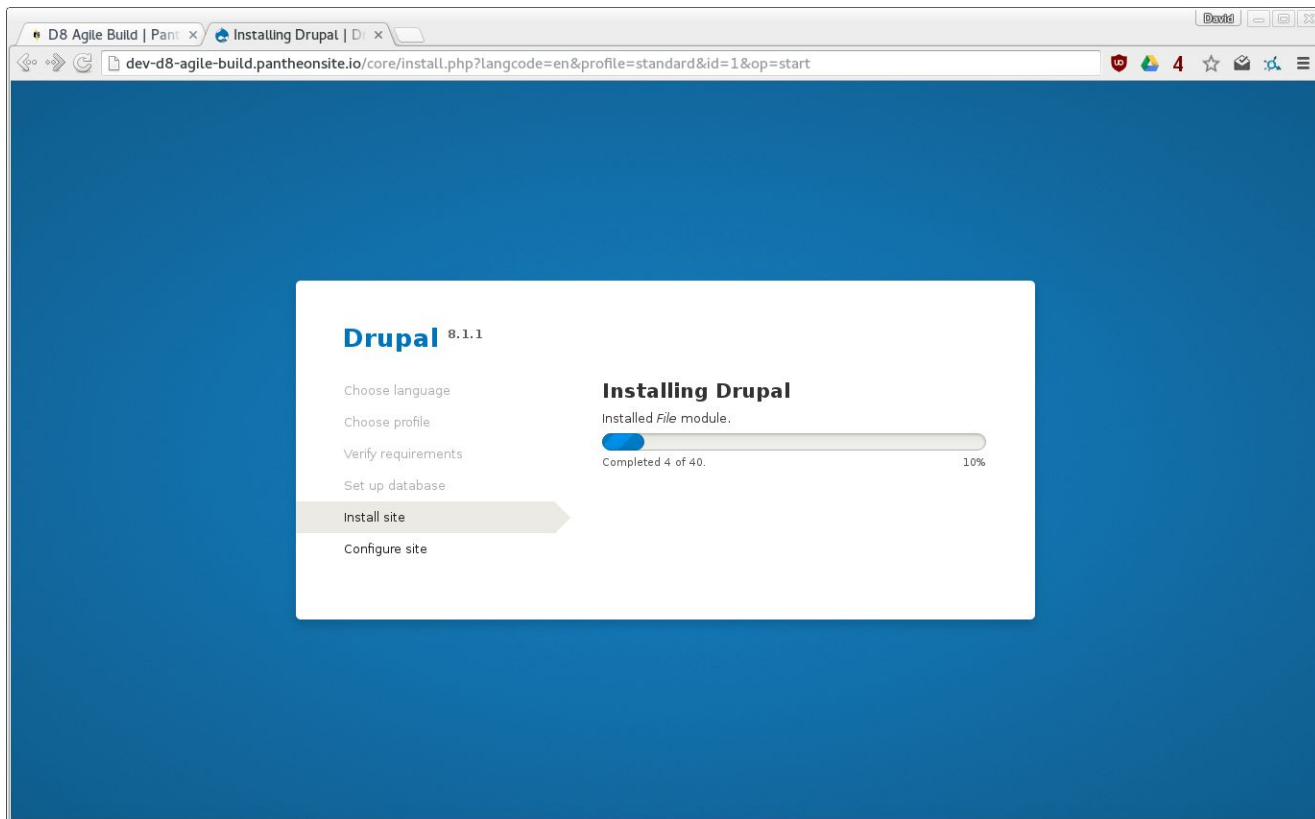


The screenshot shows the Pantheon D8 Agile Build dashboard. The top navigation bar includes the Pantheon logo, documentation, support, contact sales, and a user profile for David. The main header displays 'D8 Agile Build' with 'Pantheon Support' and 'Business' tabs. Below this, there are tabs for Multidev, Dev, Test, and Live. The left sidebar contains links for Visit Development Site, Site Admin, Code, Merge, Status, Database / Files, Errors, Domains / HTTPS, Backups, and Security. The main content area shows the 'Code' tab selected, with a 'Connection Mode' section set to 'SFTP'. Below this is a commit form for David Strauss, showing 'No changes to commit' and a 'Commit' button. A 'Commit Log' section below shows an 'Initial Commit' by Root, authored 7 minutes ago, with a commit hash of a407db8e41. A 'Help' button is located in the bottom right corner.

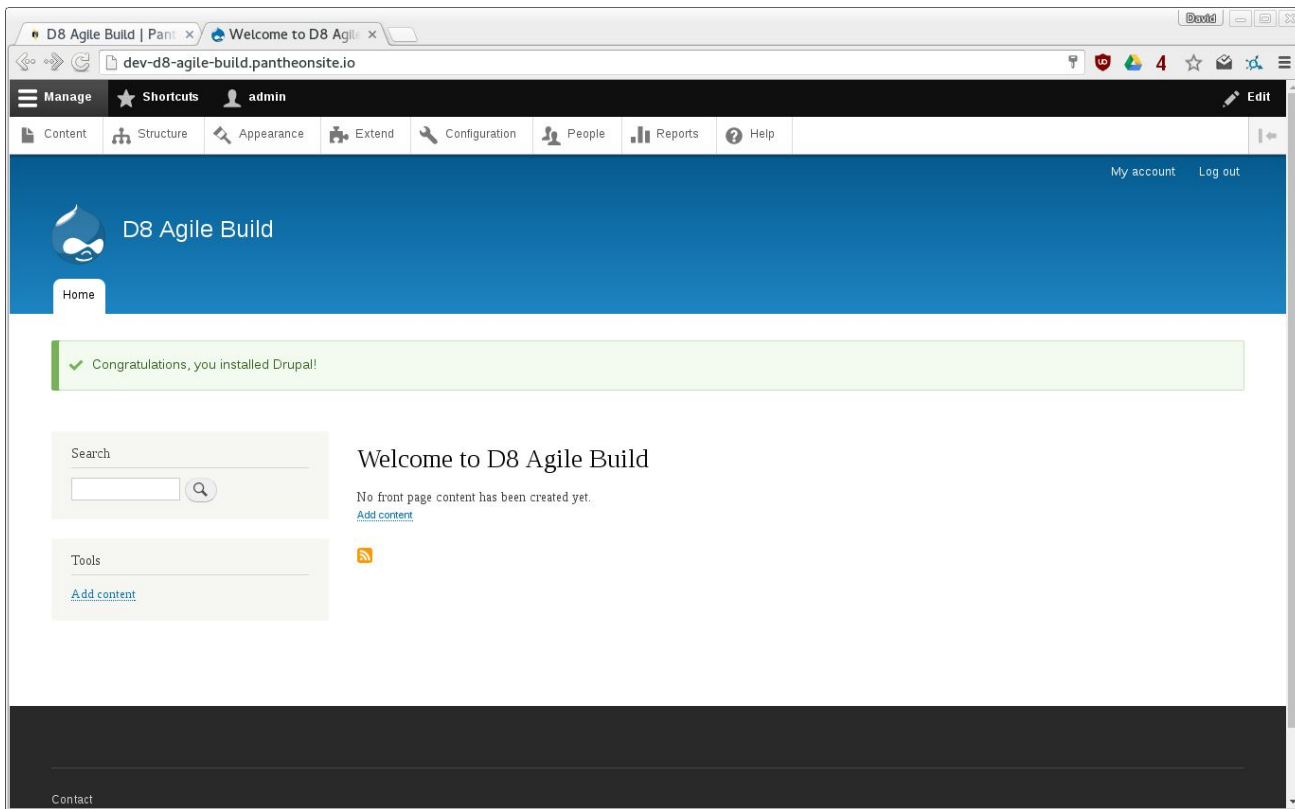
# Use the “Standard” Installation Profile



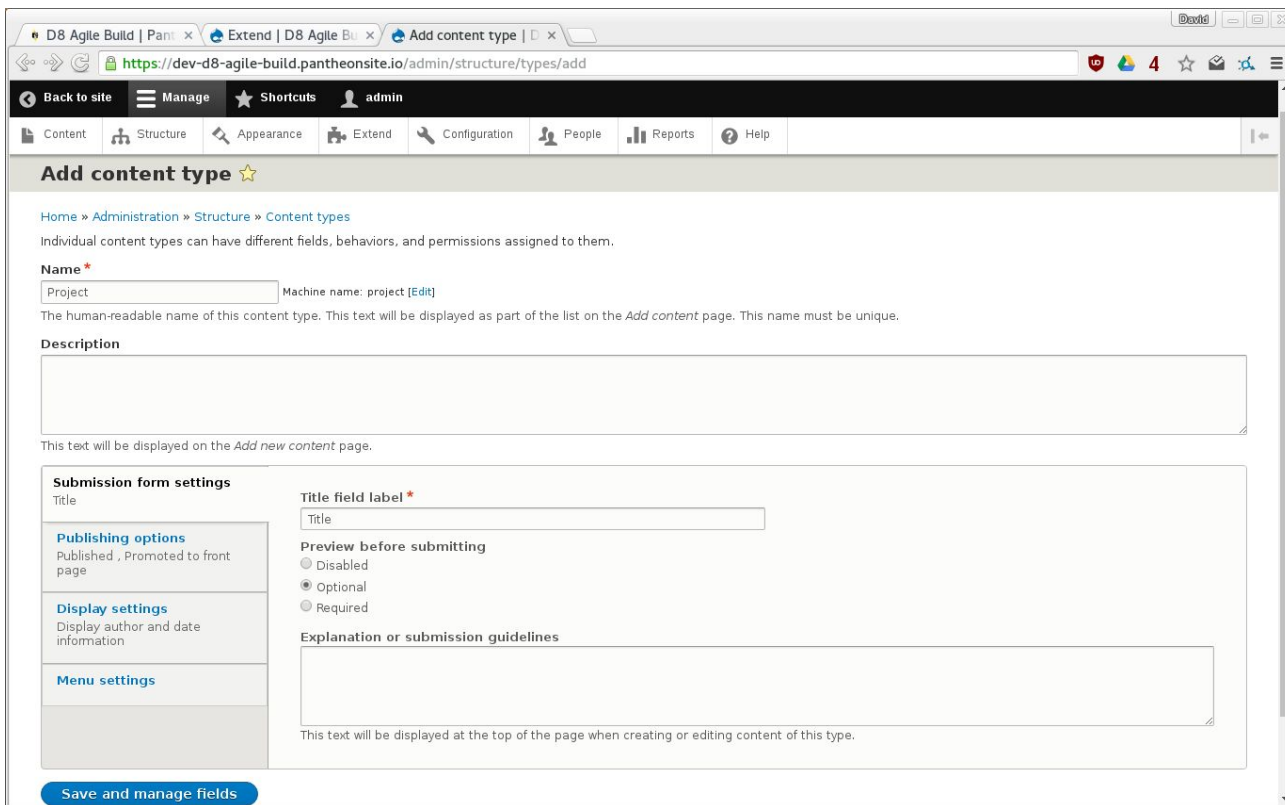
PANTHEON®



# Lookin' Good So Far



# Add a“Project” Content Type



The screenshot shows the 'Add content type' page in the Pantheon admin interface. The browser address bar shows the URL `https://dev-d8-agile-build.pantheonsite.io/admin/structure/types/add`. The page has a dark navigation bar with links for 'Back to site', 'Manage', 'Shortcuts', and 'admin'. Below this is a secondary navigation bar with icons for 'Content', 'Structure', 'Appearance', 'Extend', 'Configuration', 'People', 'Reports', and 'Help'. The main content area is titled 'Add content type' with a star icon. It includes a breadcrumb trail: 'Home » Administration » Structure » Content types'. A paragraph explains that individual content types can have different fields, behaviors, and permissions. The 'Name' field is required and contains the text 'Project'. A machine name field shows 'project' with an 'Edit' link. A description field is empty. Below the description field, a note states: 'This text will be displayed on the Add new content page.' The 'Submission form settings' section is divided into two columns. The left column has links for 'Publishing options' (Published, Promoted to front page), 'Display settings' (Display author and date information), and 'Menu settings'. The right column has a 'Title field label' field containing 'Title', a 'Preview before submitting' section with radio buttons for 'Disabled' (selected), 'Optional', and 'Required', and an 'Explanation or submission guidelines' text area. A note below the text area states: 'This text will be displayed at the top of the page when creating or editing content of this type.' At the bottom of the form is a blue button labeled 'Save and manage fields'.

**Add content type** ☆

Home » Administration » Structure » Content types

Individual content types can have different fields, behaviors, and permissions assigned to them.

**Name \***

Project Machine name: project [Edit]

The human-readable name of this content type. This text will be displayed as part of the list on the *Add content* page. This name must be unique.

**Description**

This text will be displayed on the *Add new content* page.

**Submission form settings**

Title

**Title field label \***

Title

**Preview before submitting**

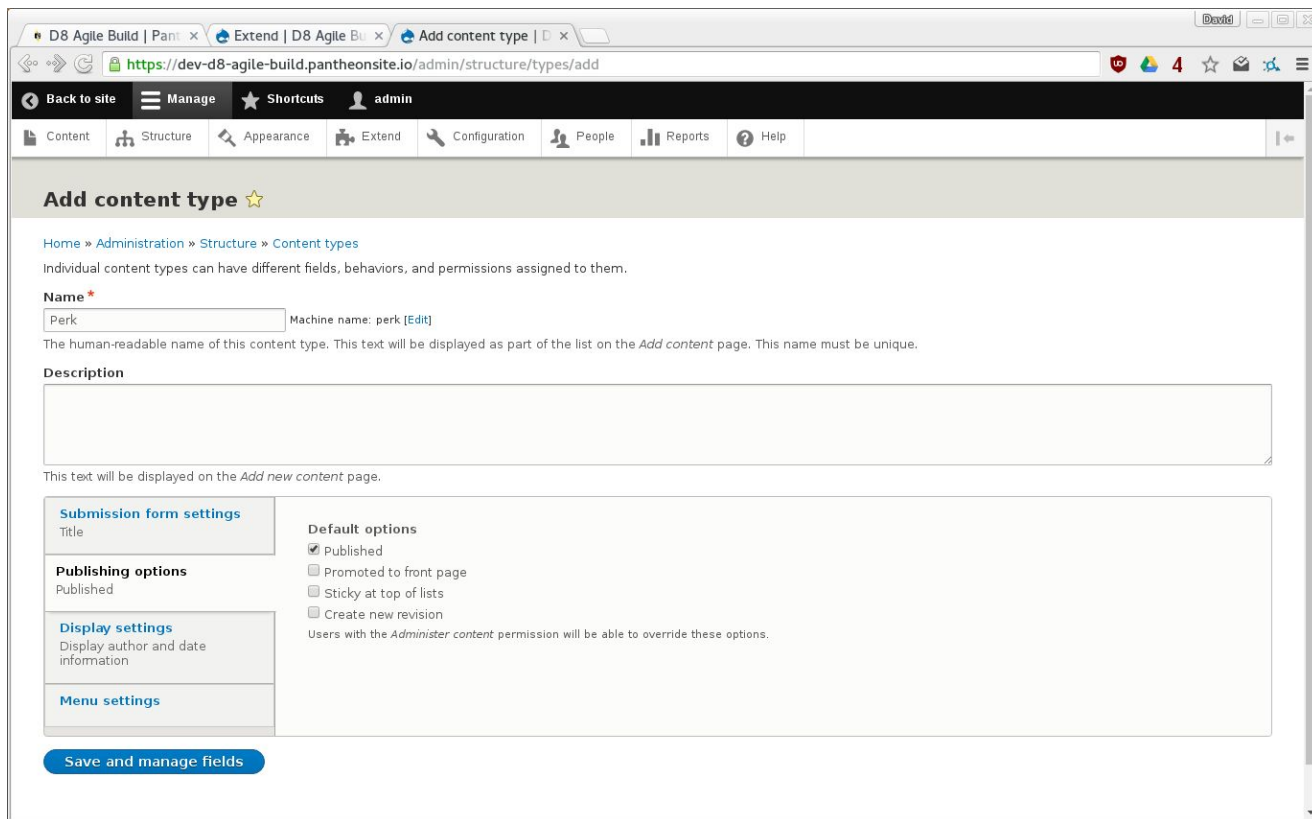
☐ Disabled  
☐ Optional  
☐ Required

**Explanation or submission guidelines**

This text will be displayed at the top of the page when creating or editing content of this type.

**Save and manage fields**

# Add a “Perks” Content Type



The screenshot shows the 'Add content type' form in a web browser. The browser tabs include 'D8 Agile Build | Panthe...', 'Extend | D8 Agile Bu...', and 'Add content type | D8 Agile Bu...'. The address bar shows the URL 'https://dev-d8-agile-build.pantheon.io/admin/structure/types/add'. The page has a dark navigation bar with links like 'Back to site', 'Manage', 'Shortcuts', and 'admin'. Below this is a secondary navigation bar with icons for 'Content', 'Structure', 'Appearance', 'Extend', 'Configuration', 'People', 'Reports', and 'Help'. The main heading is 'Add content type' with a star icon. A breadcrumb trail reads 'Home » Administration » Structure » Content types'. A paragraph explains that individual content types can have different fields, behaviors, and permissions. The 'Name' field is labeled with a red asterisk and contains the text 'Perk'. To its right, it says 'Machine name: perk [Edit]'. A note below states: 'The human-readable name of this content type. This text will be displayed as part of the list on the Add content page. This name must be unique.' The 'Description' field is a large text area. Below it, a note says: 'This text will be displayed on the Add new content page.' On the left, there are four expandable sections: 'Submission form settings' (with a 'Title' field), 'Publishing options' (with a 'Published' checkbox), 'Display settings' (with 'Display author and date information'), and 'Menu settings'. On the right, under 'Default options', there are three checkboxes: 'Published' (checked), 'Promoted to front page', and 'Sticky at top of lists'. Below these is a checkbox for 'Create new revision'. A note at the bottom of the default options section says: 'Users with the Administer content permission will be able to override these options.' At the bottom left, there is a blue button labeled 'Save and manage fields'.

**Add content type** ☆

Home » Administration » Structure » Content types

Individual content types can have different fields, behaviors, and permissions assigned to them.

**Name \***

Perk Machine name: perk [Edit]

The human-readable name of this content type. This text will be displayed as part of the list on the Add content page. This name must be unique.

**Description**

This text will be displayed on the Add new content page.

**Submission form settings**

Title

**Publishing options**

Published

**Display settings**

Display author and date information

**Menu settings**

**Default options**

☒ Published

☐ Promoted to front page

☐ Sticky at top of lists

☐ Create new revision

Users with the Administer content permission will be able to override these options.

**Save and manage fields**

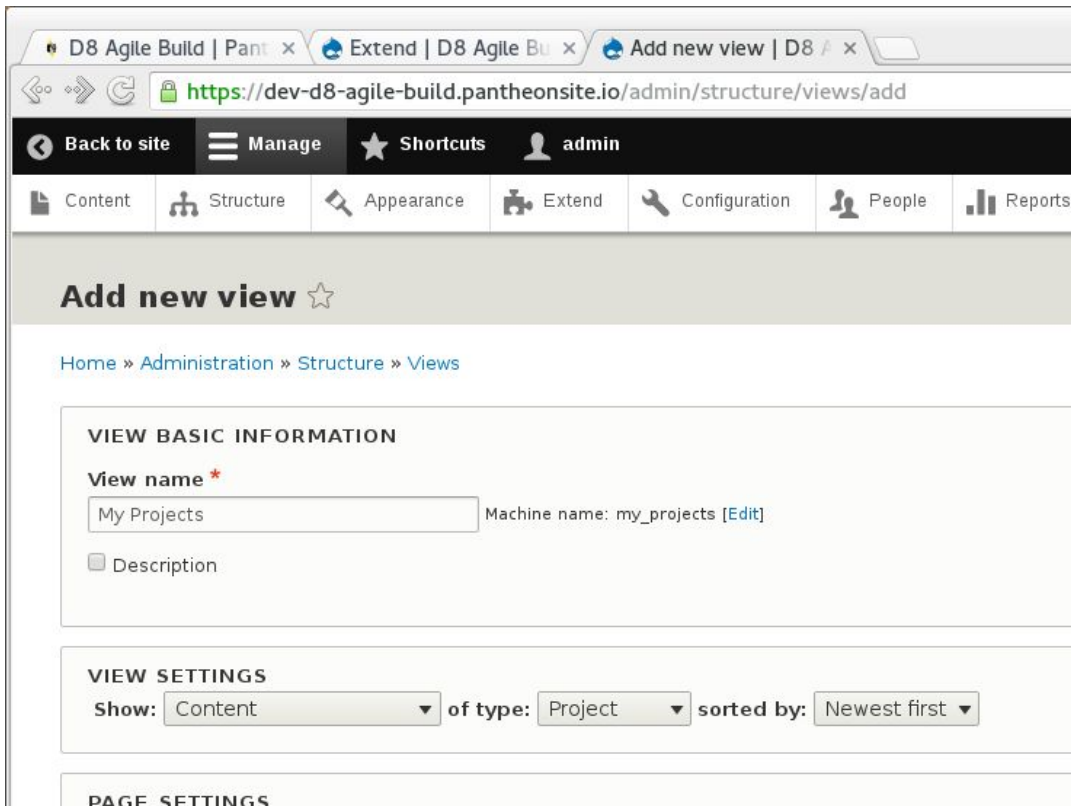
## Done

- Installation
- Content Types
  - ◉ Projects
  - ◉ Perks

## To Do

- Link Perks to Projects
  - ◉ And list them on the Project page
- Accept payment for Perks
- Support videos on Projects

# Create a View to List Projects...



The screenshot shows the Pantheon admin interface for creating a new view. The browser tabs include "D8 Agile Build | Pant...", "Extend | D8 Agile Bu...", and "Add new view | D8 / x". The address bar shows the URL "https://dev-d8-agile-build.pantheonsite.io/admin/structure/views/add". The navigation bar includes "Back to site", "Manage", "Shortcuts", and "admin". Below the navigation bar are tabs for "Content", "Structure", "Appearance", "Extend", "Configuration", "People", and "Reports". The main heading is "Add new view" with a star icon. The breadcrumb trail is "Home » Administration » Structure » Views". The form is divided into three sections: "VIEW BASIC INFORMATION", "VIEW SETTINGS", and "PAGE SETTINGS".

**VIEW BASIC INFORMATION**

**View name \***

My Projects Machine name: my\_projects [Edit]

☐ Description

**VIEW SETTINGS**

Show: Content of type: Project sorted by: Newest first

**PAGE SETTINGS**

# ...and Give It a Conditional Filter by User



PANTHEON®



## My Projects

Home » Administration

## Displays

Master\* + Add

### TITLE

Title: None

### FORMAT

Format: Unformatted

Show: Fields |

### FIELDS

Content: Title

### FILTER CRITERIA

Content: Publishing

Content: Content

### SORT CRITERIA

Content: Authored

## Configure contextual filter: Content: Authored by

☒ Provide default value

### Type

User ID from logged in user ▼

☐ Hide view

☐ Display a summary

☐ Display contents of "No results found"

☐ Display "Access Denied"

### ▶ EXCEPTIONS

☐ Skip default argument for view URL

Select whether to include this default argument when constructing the URL for this view. Skipping default arguments is useful e.g. in the case of feeds.

### ▼ WHEN THE FILTER VALUE IS AVAILABLE OR A DEFAULT IS PROVIDED

☐ Override title

☒ Specify validation criteria

### Validator

User ID ▼

☒ Validate user has access to the User

### Access operation to check

☐ View

☒ Edit

# ...and Make the View an Entity Reference



PANTHEON®



## My Projects (Content) ☆

[Home](#) » [Administration](#) » [Structure](#) » [Views](#)

✓

The view *My Projects* has been saved.

⚠

You have unsaved changes.

### Displays

Master\*

+ Add

Attachment

Block

Embed

Entity Reference

Feed

Page

|                |                  |         |
|----------------|------------------|---------|
| TITLE          |                  | Access  |
| Title: Non     |                  | HEADER  |
| FORMAT         |                  | FOOTER  |
| Format: U      | ings             | NO RES  |
| Show: Fields   | Settings         | PAGER   |
| FIELDS         | <div>Add ▼</div> | Use pa  |
| Content: Title |                  | More li |

# Entity References Require a “Search Field”



PANTHEON®



Entity Reference: Style options

Search fields \*

☒ Content: Title

Select the field(s) that will be searched when using the autocomplete widget.

Apply Cancel

NO RESULTS BEHAVIOR

# Now We Can Have Perks Linked to Projects



PANTHEON®



D8 Agile Build | Pant x Extend | D8 Agile Bu x Add field | D8 Agile E x

https://dev-d8-agile-build.pantheonsite.io/admin/structure/types/manage/perk/fie

Back to site Manage Shortcuts admin

Content Structure Appearance Extend Configuration People Re

## Add field ☆

Home » Administration » Structure » Content types » Perk » Manage fields

**Add a new field** **Re-use an existing field**

Content or - Select an existing field -

**Label\***

Project Machine name: field\_project [Edit]

Save and continue

# ...and Let Users Pick from Their Projects



PANTHEON®



## ▼ REFERENCE TYPE

### Reference method \*

Views: Filter by an entity reference view ▼

### View used to select the entities \*

my\_projects - Entity Reference ▼

Choose the view and display that select the entities that can be referenced.  
Only views with a display of type "Entity Reference" are eligible.

### View arguments

Provide a comma separated list of arguments to pass to the view.

Save settings

[Delete](#)

# Create a Block Showing Perks for a Project



PANTHEON®



David

D8 Agile Build | Pant x Extend | D8 Agile Bu x Add new view | D8 A x

https://dev-d8-agile-build.pantheonsite.io/admin/structure/views/add

Back to site Manage Shortcuts admin

Content Structure Appearance Extend Configuration People Reports Help

## Add new view ☆

Home » Administration » Structure » Views

### VIEW BASIC INFORMATION

**View name \***

Perks by Project Machine name: perks\_by\_project [Edit]

☐ Description

### VIEW SETTINGS

Show: Content of type: Perk sorted by: Newest first

# ...Filtered by Project



PANTHEON®



## Perks by Project

Home » Administration

You have unsaved changes

### Displays

Block\* + Add

Display name: Perk

#### TITLE

Title: None

#### FORMAT

Format: Uniform

Show: Fields

#### FIELDS

Content: Title

#### FILTER CRITERIA

Content: Publishing

Content: Content type (= Perk)

### Add contextual filters



#### Search

#### Type

- All -

|                                     | TITLE                   | CATEGORY | DESCRIPTION                               |
|-------------------------------------|-------------------------|----------|-------------------------------------------|
| <input type="checkbox"/>            | Body (body)             | Content  | Appears in: article, page, project, perk. |
| <input type="checkbox"/>            | Body (body:format)      | Content  | Appears in: article, page, project, perk. |
| <input type="checkbox"/>            | Body (body:summary)     | Content  | Appears in: article, page, project, perk. |
| <input type="checkbox"/>            | Preview (field_preview) | Content  | Appears in: project.                      |
| <input checked="" type="checkbox"/> | Project (field_project) | Content  | Appears in: perk.                         |

Selected: Project (field\_project)

Add and configure contextual filters

Cancel

# ...for the Current Content View



PANTHEON®



This display does not have a source for contextual filters, so no contextual filter value will be available unless you select 'Provide default'.

## ▼ WHEN THE FILTER VALUE IS *NOT* AVAILABLE

- ☐ Display all results for the specified field
- ☒ Provide default value

### Type

Content ID from URL ▼

- ☐ Hide view
- ☐ Display a summary
- ☐ Display contents of "No results found"
- ☐ Display "Access Denied"

## ► EXCEPTIONS

# ...and Place It in the Right Sidebar



Back to site Manage Shortcuts admin

Content Structure Appearance Extend Configuration People Reports Help

Sidebar first Place block

|          |       |                 |             |
|----------|-------|-----------------|-------------|
| ✚ Search | Forms | Sidebar first ▼ | Configure ▼ |
| ✚ Tools  | Menus | Sidebar first ▼ | Configure ▼ |

Sidebar second Place block

*No blocks in this region*

Featured bottom first Place block



# ...and Place It in the Right Sidebar



PANTHEON®



Place block

+ Add custom block

Filter by block name

| BLOCK                 | CATEGORY      | OPERATIONS  |
|-----------------------|---------------|-------------|
| Page title            | core          | Place block |
| Primary admin actions | core          | Place block |
| Tabs                  | core          | Place block |
| Search form           | Forms         | Place block |
| User login            | Forms         | Place block |
| Help                  | Help          | Place block |
| Perks by Project      | Lists (Views) | Place block |
| Recent comments       | Lists (Views) | Place block |

# ...Restricted to Project Pages



PANTHEON®



Configure block

Block description: Perks by Project

☒ Display title

Items per block

10 (default setting) ▼

☐ Override title

Visibility

Content types

Project

Pages

Not restricted

Roles

Not restricted

Content types

☐ Article

☐ Basic page

☐ Perk

☒ Project

Machine-readable name \*

## Done

- Installation
- Content Types
  - ◉ Projects
  - ◉ Perks
- Link Perks to Projects
  - ◉ And list them on the Project page

## To Do

- Accept payment for Perks
- Support videos on Projects

# Add in Modules for Video and Payment



PANTHEON®



straussd@titan:~/Projects/d8-agile-build

File Edit View Search Terminal Help

```
[straussd@titan d8-agile-build]$ php ~/drush.phar dl video_embed_field
Project video_embed_field (8.x-1.0-rc8) downloaded to [success]
/home/straussd/Projects/d8-agile-build/modules/video_embed_field.
Project video_embed_field contains 3 modules: video_embed_wysiwyg, video_embed_media, video_embed_field.
[straussd@titan d8-agile-build]$ php ~/drush.phar dl payment
Project payment (8.x-2.0-rc2) downloaded to [success]
/home/straussd/Projects/d8-agile-build/modules/payment.
Project payment contains 5 modules: payment_form, payment_test, payment_reference, payment_reference_test, payment.
```

# Add a Video Preview to Projects



PANTHEON®



Browser tabs: D8 Agile Build | Pant x, Extend | D8 Agile Bu x, Add field | D8 Agile E x

URL: <https://dev-d8-agile-build.pantheonsite.io/admin/structure/types/manage/project/>

Navigation: Back to site, Manage, Shortcuts, admin

Menu: Content, Structure, Appearance, Extend, Configuration, People, Rep

## Add field ☆

[Home](#) » [Administration](#) » [Structure](#) » [Content types](#) » [Project](#) » [Manage fields](#)

**Add a new field** **Re-use an existing field**

Video Embed or - Select an existing field -

**Label \***

Preview Machine name: field\_preview [Edit]

[Save and continue](#)

# ...and Configure the Video Field



PANTHEON®



✓ Updated field *Preview* field settings.

## Label \*

Preview

## Help text

Instructions to present to the user below this field on the editing form.

Allowed HTML tags: <a> <b> <big> <code> <del> <em> <i> <ins> <pre> <q> <small>

This field supports tokens.

☐ Required field

## ▼ DEFAULT VALUE

The default value for this field, used when creating new content.

## Preview

## Allowed Providers

☐ YouTube Playlist

☐ Vimeo

☒ YouTube

Restrict users from entering information from the following providers. If none are selected any video

Save settings

Delete

# Add a Payment Form Field to Perks



PANTHEON®



D8 Agile Build | Pant x Extend | D8 Agile Bu x Add field | D8 Agile E x

https://dev-d8-agile-build.pantheonsite.io/admin/structure/types/manage

Back to site Manage Shortcuts admin

Content Structure Appearance Extend Configuration People

## Add field ☆

Home » Administration » Structure » Content types » Perk » Manage fields

Add a new field Re-use an existing field

Payment form or - Select an existing field -

Label \*  
Payment Machine name: field\_payment [Edit]

Save and continue

# ...and Complete Its Configuration



PANTHEON®



[Back to site](#) [Manage](#) [Shortcuts](#) [admin](#)

[Content](#) [Structure](#) [Appearance](#) [Extend](#) [Configuration](#)

✓ Updated field *Payment* field settings.

**Label \***

**Help text**  

Instructions to present to the user below this field on the editing form.  
Allowed HTML tags: <a> <b> <big> <code> <del> <em> <i> <ins> <pre> <q> <sr>  
This field supports tokens.

☒ Required field

**▼ DEFAULT VALUE**

The default value for this field, used when creating new content.

| NAME                         | TYPE | CONFIGURATION |
|------------------------------|------|---------------|
| There are no line items yet. |      |               |

**Type**

# Simplify the Form for Users



PANTHEON®



## Manage form display ☆

[Edit](#)[Manage fields](#)[Manage form display](#)[Manage display](#)

[Home](#) » [Administration](#) » [Structure](#) » [Content types](#) » [Perk](#)

Content items can be edited using different form modes. Here, you can define which fields are shown and hidden when *Perk* content is edited in each form mode, and are displayed in each form mode.

| FIELD                    | WIDGET                     |                                                |
|--------------------------|----------------------------|------------------------------------------------|
| ✚ Title                  | Textfield ▼                | Textfield size: 60                             |
| ✚ Body                   | Text area with a summary ▼ | Number of rows: 9<br>Number of summary rows: 3 |
| ✚ Payment                | Drop-down selection list ▼ |                                                |
| <b>Disabled</b>          |                            |                                                |
| ✚ Authored by            | - Hidden - ▼               |                                                |
| ✚ Authored on            | - Hidden - ▼               |                                                |
| ✚ Promoted to front page | - Hidden - ▼               |                                                |
| ✚ Sticky at top of lists | - Hidden - ▼               |                                                |
| ✚ URL alias              | - Hidden - ▼               |                                                |

# Final “Project” Form



[Back to site](#) [Manage](#) [Shortcuts](#) [admin](#)

[Content](#) [Structure](#) [Appearance](#) [Extend](#) [Configuration](#) [People](#) [Reports](#) [Help](#)

## Create Project ☆

[Home](#) » [Add content](#)

**Title \***

**Preview**

**Body (Edit summary)**

**B** **I**      Normal 

Description

body p

**Text format** Basic HTML [About text formats ?](#)

**Risks \***

**B** **I**      Normal 

I've never done this before.

**Last saved:** Not saved yet

**Author:** admin

☐ Create new revision

► MENU SETTINGS

► URL PATH SETTINGS

# Final “Perk” Form



[Back to site](#) [Manage](#) [Shortcuts](#) [admin](#)

[Content](#) [Structure](#) [Appearance](#) [Extend](#) [Configuration](#) [People](#) [Reports](#) [Help](#)

[Home](#) » [Add content](#)

**Title \***

**Body (Edit summary)**  

**B** **I**       **Format**  **Source**

- Free [Ebook](#)
- Coffee with the Developers

body ul li

**Text format** **Basic HTML** [About text formats ?](#)

**Amount \***  

**U.S. dollar (USD)**

**Quantity \***

**Description \***

**Project**

[Save and publish](#) [Preview](#)

**Last saved:** Not saved yet  
**Author:** admin  
☐ Create new revision  
**► MENU SETTINGS**

# Time to Theme and Trim Down the Display



PANTHEON®





## D8 Agile Build

Home

Home

Search

Q

### My New Project

Submitted by admin on Mon, 05/09/2016 - 19:29

**Preview**



⌚ ↗

Description

**Risks**

I've never done this before.

#### My New Perk

Submitted by admin on Mon, 05/09/2016 - 19:31

- Free Ebook
- Coffee with the Developers

| Description  | Quantity | Amount     | Total      |
|--------------|----------|------------|------------|
| My New Perk  | 1        | USD 100.00 | USD 100.00 |
| Total amount | XXX      | 100.000000 |            |

**Payment method**

There are no available options.

Pay

# Finished Product!



 Fund Everything

HomeCategoriesProjects

## Funding Your Dreams

Let's make a world where everyone gets everything they want.

### Welcome to Agile D8 Builds: Doing the Most Without PHP

Submitted by matt on Wed, 05/04/2016 - 22:41

One of the design philosophies behind Drupal 8 is the ability to make simple things easy and complex things possible. While there is a lot of neat stuff "under the hood" with Drupal 8, this talk will discuss how you can leverage the Drupal 8 core stack to build pretty awesome websites without writing (or including) a line of custom or contributed PHP.

[Read more](#) [Log in](#) or [register](#) to post comments

#### Featured Projects

[Bring Catalyst to Drupalcon](#)

*This is a project to bring Catalyst, one of the great Djs of our time, to Drupalcon.*

[Throw an UFO Alien Dance Party](#)

*This is a project to call all occupants of interplanetary craft for a dance party.*

[Create a Share Our Wealth Club](#)

*This is a project to make every one a king, but with no one wearing a crown.*

[Let's Go Back to Bali](#)

*This is a project to make a return trip to Bali to find love, adventure, and champagne.*









#### Search Our Projects

Look into your heart and find the thing you most want. Enter it into the search box and hit the enter key. Your hopes and dreams should soon appear.

Search

#### About the Site

This site is a demonstration of one of the design philosophies behind Drupal 8 which is the ability to make simple things easy and complex things possible. While there is a lot of neat stuff "under the hood" with Drupal 8, this site will help explain how you can leverage the Drupal 8 core stack to build pretty awesome websites without writing (or including) a line of custom or contributed PHP.

 Drupal

# **Closing Thoughts & QA**